# STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SMOOTHING-AT-LCI-POINT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3292-3305 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3307-3326, the complete author proof (18 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 26 of queue_examples-defos.json (pb=1082, stmt 3292-3305, proof 3307-3326).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 00SD: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3326–3326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 00P1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3326–3326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 00SF: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3326–3326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 00SY: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3326–3326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0C3K: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3326–3326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0E7V: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3216–3221. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $k$ be a field. Let $A$ be a $k$-algebra. Assume*

(1) *$A$ is a local ring essentially of finite type over $k$,*
(2) *$A$ is a complete intersection over $k$ (Algebra, Definition 00SD).*

*Set $d = \dim(A) + trdeg_k(\kappa)$ where $\kappa$ is the residue field of $A$. Then there exists an integer $n$ and a flat, essentially of finite type ring map $k[[t]] \to B$ with $B/tB \cong A$ such that $t^n$ is in the $d$th Fitting ideal of $\Omega_{B/k[[t]]}$.*

*Proof.* By Algebra, Lemma 00SF we can write $A$ as the localization at a prime $\mathfrak{p}$ of a global complete intersection $P$ over $k$. Observe that $\dim(P) = d$ by Algebra, Lemma 00P1. By Lemma 0E7V we can find a flat, finite type ring map $k[[t]] \to Q$ such that $P \cong Q/tQ$ and such that $k((t)) \to Q[1/t]$ is smooth. It follows from the construction of $Q$ in the lemma that $k[[t]] \to Q$ is a relative global complete intersection of relative dimension $d$; alternatively, Algebra, Lemma 00SY tells us that $Q$ or a suitable principal localization of $Q$ is such a global complete intersection. Hence by Divisors, Lemma 0C3K the $d$th Fitting ideal $I \subset Q$ of $\Omega_{Q/k[[t]]}$ cuts out the singular locus of $\mathrm{Spec}(Q) \to \mathrm{Spec}(k[[t]])$. Thus $t^n \in I$ for some $n$. Let $\mathfrak{q} \subset Q$ be the inverse image of $\mathfrak{p}$. Set $B = Q_{\mathfrak{q}}$. The lemma is proved. □